

Teaching Exam

Subject – Computer Science

Topic – DBMS

Lecture No. – 4



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Topics to be Covered

Topic



- 1:- Requirement of Keys
- 2:- Super key and it's Feature
- 3:- Candidate Key and it's Feature
- 4:- Primary key and it's Feature
- 5:- Foreign key
- 6:- Composite Key and it's Feature
- 7:- Alternate Key
- 8:- Relational Model ✓
- 9:- Practice On MCQ Problem solving with ✓
Discussion.



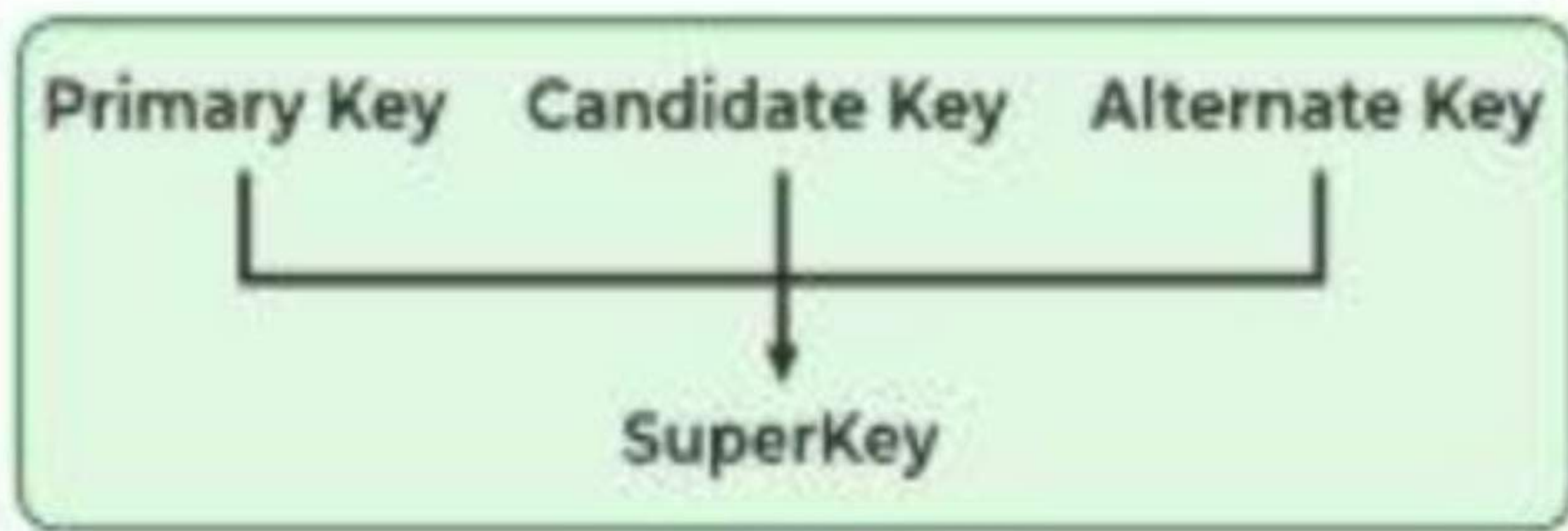
Topic : Requirement of Keys

1:-We require keys in a DBMS to ensure that data is organized, accurate, and easily accessible. Keys help to uniquely identify records in a table, which prevents duplication and ensures data integrity.

2:-Keys also establish **relationships between different tables, allowing for efficient querying and management of data.** Without keys, it would be difficult to retrieve or update specific records, and the database could become inconsistent or unreliable.



Topic : Super Key





Topic : Super Key and it's Features

A **Super Key** in DBMS is a group of one or more attributes in a table that can uniquely identify every row in that table. It ensures no two rows have the same combination of values for those attributes.

Key Features of Super Key in DBMS:-

Uniqueness: ✓

A super key ensures that every row in a table is uniquely identified and prevents duplicate records.

One or More Attributes: ✓

It can consist of a single column or a combination of multiple columns.

Not Necessarily Minimal: ✓

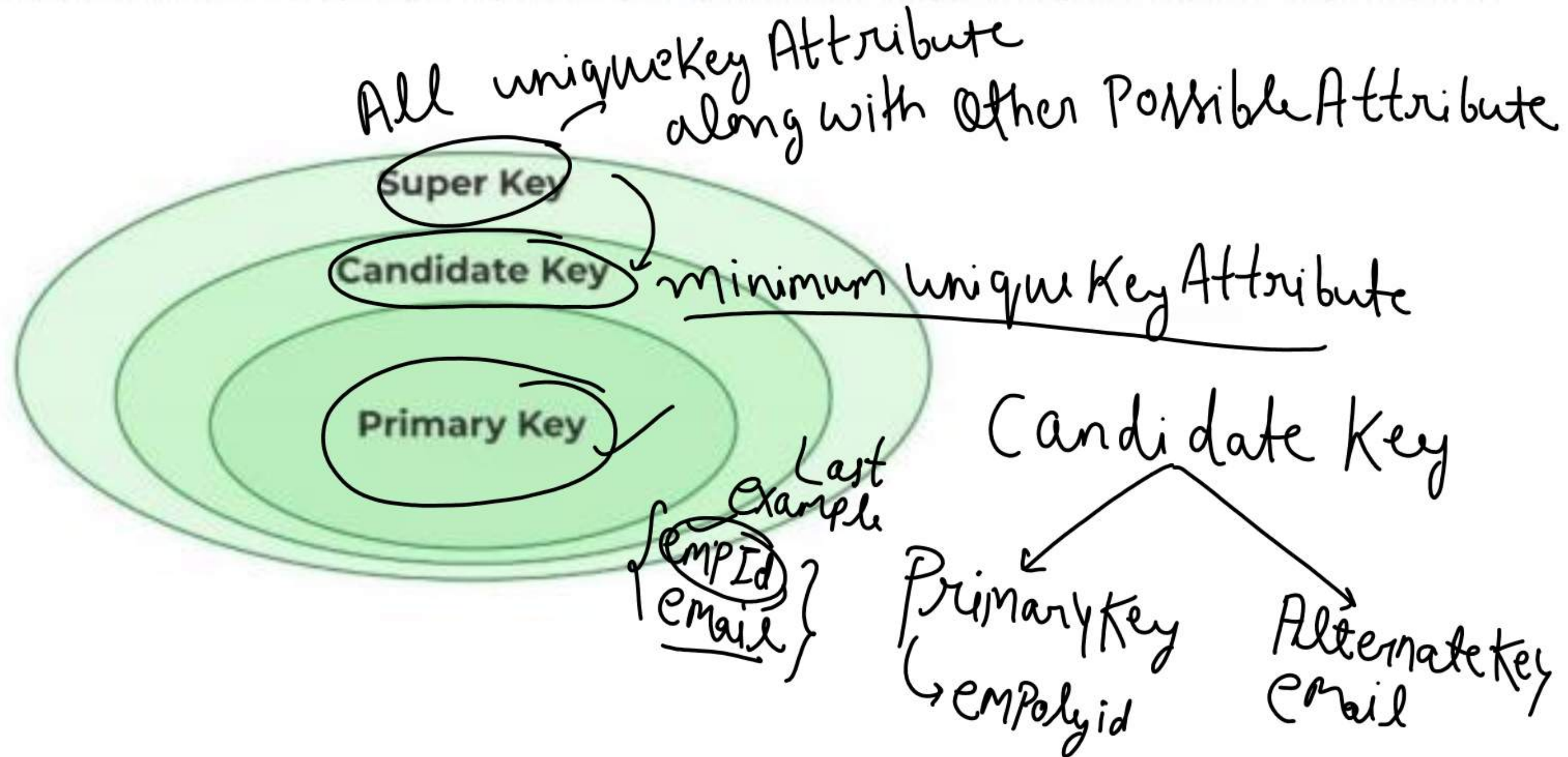
A super key may include extra attributes that are not required for uniqueness, unlike a candidate key.



Topic : Super Key and it's Features

Includes Primary Key and Candidate Keys:

Every primary key and candidate key is a super key, but not all super keys are primary keys or candidate key.





Topic : Super Key

Examples of Super Key in DBMS with Tables

Let's consider a table called **Employees**:

Employee_ID	Name	Department	Email
101	Alice	HR	alice@company.com
102	Bob	IT	bob@company.com
103	Charlie	IT	charlie@company.com

In this example:

- **Employee_ID** uniquely identifies each row in the table.
- Any combination of attributes that includes **Employee_ID** will also be a super key because **Employee_ID** alone is sufficient for uniqueness.



Topic : Super Key

So, from the above table, we can conclude the following set of the super keys:

- {Employee_ID}
- {Employee_ID, Name}
- {Employee_ID, Department}
- {Employee_ID, Email}
- {Employee_ID, Name, Department}
- {Employee_ID, Name, Email}
- {Employee_ID, Department, Email}
- {Employee_ID, Name, Department, Email}



Topic : Candidate Key and it's Feature

A candidate key is a minimal set of attributes that uniquely identifies each tuple within a table. In other words, there should not be any two rows in a table that can have the same values for the columns that are the part of candidate key. **Candidate key is a subset of Super key.** It is very important for establishing relationships between tables and maintaining data integrity.

Key Features of Candidate Key in DBMS:-

- A candidate key has unique values for every row in a table.
- Every candidate key is a super key, but not all super keys qualify as candidate keys. Simply put, a candidate key is a minimal super key.
- A candidate key can contain null values, but this depends on the database's design and constraints.



Topic : Feature of Candidate Key

- It should not have redundant attributes, meaning it must be as simple as possible to uniquely identify a row.
- Using a candidate key, we can uniquely determine the values of all other attributes in a table.
- The primary key is chosen from the set of candidate keys.
- Candidate keys help us classify table attributes into:
 - **Prime Attributes:** Attributes that form part of the candidate key.
 - **Non-Prime Attributes:** Attributes not included in any candidate key.
- Candidate keys ensure data integrity and prevent duplication in the table.



Topic : Example of Candidate Key

Example of Candidate Key

Let's try to understand, the concept of the candidate key with an example of a student table.

Candidate Key			
Candidate Key		Candidate Key	
Roll No.	Name	Age	Phone
1	Nick	21	8736283921
2	John	25	7378348749
3	Case	20	7648196483
4	Dawid	21	9349783479

Candidate Key



Topic : Example of Candidate Key

Here's how the candidate keys are identified from the table:

1. Roll No.:

- Each Roll No. is unique for every student.
- It can uniquely identify each row in the table.
- Hence, Roll No. is a **candidate key**.

2. Phone:

- Each Phone number is also unique for every student.
- It can also uniquely identify each row in the table.
- Hence, Phone is another **candidate key**.



Topic : PrimaryKey and it's Feature

A primary key in a table that uniquely identifies each row and column or set of columns in the table. The primary key is an attribute or a set of attributes that help to uniquely identify the tuples(records) in the relational table.

primary key is used to identify the tuples of a relation none of its attribute values can be null.



Topic : PrimaryKey and it's Feature

Rules For Defining a Primary Key

- **Minimal:** The primary key is composed of a minimum number of attributes that satisfy the unique occurrence of the tuples or records. So if one candidate key is formed using two attributes and the other using a single attribute then the one with a single attribute key should be chosen as the primary key.
- **Accessible:** The primary key is used to check the ability to access and interact with the database. The user must easily create, read or delete a tuple using it.
- **NON NULL Value:** The primary key which refers to a nonnull value that is required for the identification of the tuple (record) means that any attribute can not contain the non-null value in DBMS.
- **Time Invariant:** The values of the primary key must not change or become null during the lifetime of the relation. For example, we cannot choose the name attribute of the EMPLOYEE Relation as a primary key because there is no guarantee that two employees with the same name will never exist.
- **Unique:** The primary key in which value can not be duplicated in any of the rows or tuples of the relation.



Topic : PrimaryKey and it's Feature

EMP_ID	Name	PAN_NO	Salary	Dept
E1	Ram	Null	23000	D1
E2	Sita	PSJ1234	Null	D2
E3	Sham	DJF2324	12000	D3
Null	Sapna	WCD232	13000	D4

Primary $\left\{ \begin{array}{l} \text{Uniqueness} \\ \text{Not Null} \end{array} \right.$

Here, we find that Employee E1 has not been yet allotted a PAN number i.e. it does not exist for him so it has been allotted a value null and will be allotted a value later on. Employee E2's Salary has not been fixed so it is unknown. Therefore we have allotted it a Null Value. But it will not cause a problem since it is not a primary key But if we allot a null value to the primary key EMP_ID then it may cause a problem as per the condition a key to be a primary key must not have a null value.

Now suppose that every employee has been allotted a PAN _NO.



Topic : PrimaryKey and it's Feature

EMP_ID	Name	PAN_NO	Salary
E1	Ram	Null	23000
E2	Sita	PSJ1234	15000
E3	Sham	DJF2324	12000
E4	Sapna	WCD232	13000

Here the attributes EMP_ID, PAN_NO, and Name (where no names repeat) can act as candidate keys for the EMPLOYEE Relation.

The Name is a time-variant attribute as when the database grows two employee may have the same names. So it may act as a possible candidate key when databases are very small. Since we know that the primary key is formed from one of the candidate keys both EMP_ID and PAN_NO can act as a primary key, but one at a time. We assume EMP_ID as the primary key in most of the cases.

A Primary Key can be formed from the combination of more than one attribute.



Topic : Foreign key and it's Feature

Foreign Key

A foreign key is an attribute in one table that refers to the **primary key** in another table.

The table that contains the foreign key is called the referencing table, and the table that is referenced is called the referenced table.

- A **foreign key** in one table points to the **primary key** in another table, establishing a relationship between them.
- It helps **connect two or more tables**, enabling you to create relationships between them. This is essential for maintaining data integrity and preventing data redundancy.
- They act as a cross-reference between the tables.
- For example, DNO is a primary key in the DEPT table and a non-key in EMP

Create database db;
Use db;

Insert into table empolyee values ('E101', 'Rahul', 'xyz@gmail.com');
Create table employee (
Employeeid varchar(30) Primary Key,

Insert into table employee values ('E101', 'Sweety', 'pqr@gmail.com');
Name Char(30),
email Varchar(20);



Topic : Foreign key and it's Feature

Primary Key

Department		
EmpNo	EmpName	DepNo
1001	Sahil	101
1004	Kavish	102
1006	Aditya	103
1005	Atul	104

Foreign Key

Foreign Key

referenced
Table

It Can be NULL

Relationship

Employee

Primary Key

Primary
Key

DepNo	DName	Location
101	HR	Delhi
102	Sales	Bangalore
103	Marketing Executive	Hyderabad
104	Technical Engineer	Chennai

Referencing
Table



Topic : Foreign key and it's Feature

Primary Key

Foreign Key

ID	Name	Course
2041	Tom	Java
2204	John	C++
2043	Alice	Python
2032	Bob	Oracle

Student Details

ID	Marks
2041	65
2204	55
2043	73
2032	62

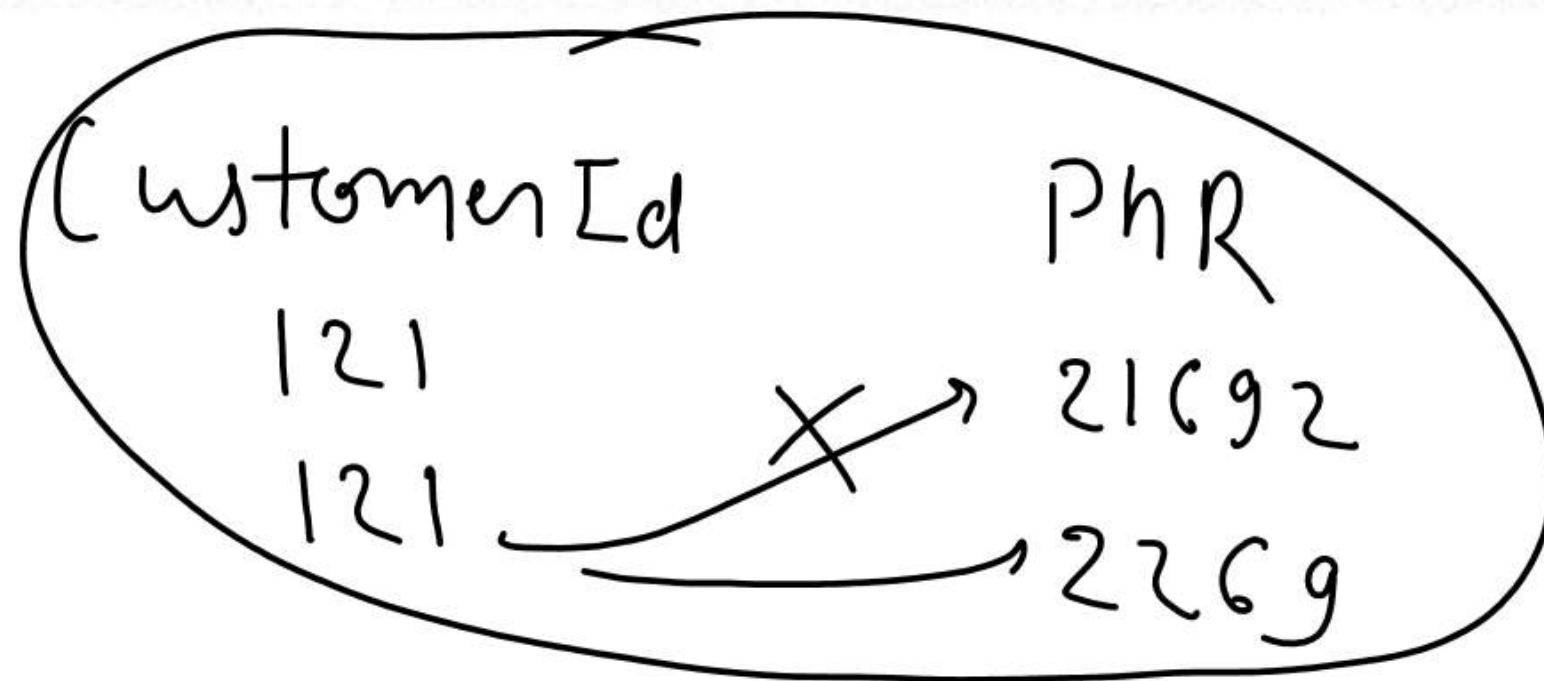
Student Marks



Topic : Composite key

A composite key is a candidate key consisting of two or more attributes (table columns) that uniquely identify a record/tuple (table row) or a relation. A compound key is a composite key for which each attribute that makes up the key is a foreign key in its own right.

- Make sure that each record has a unique identification by combining specific attributes.





Topic : Composite key

Consider we have a table *Orders* with three attributes *Customer id*, *Product id*, *Quantity*. The key

<u>Customer id</u>	<u>Product id</u>	<u>Quantity</u>
<u>121</u>	<u>9034</u>	<u>50</u>
333	9446	67
121	9869	30
201	9446	67
223	9023	80

Fig.1 Orders table

Customer id needs to be entered every time the order is placed, hence *Customer id* can repeat in the table and can not serve as a primary key. Also, *Product id* and *Quantity* cannot be chosen as primary key because multiple customers can buy same product and the same quantity. Here any of the three attributes can not be used as primary key, hence combination of these attributes need to be used.



Topic : Composite key

For example, *[Customer id + Product id]* can be used to identify the records uniquely.

Composite Key		
Customer Id	Product Id	Quantity
121	9034	50
333	9446	67
121	9869	30
201	9446	67
223	9023	80

Handwritten notes:

- Arrows point from "Customer Id" and "Product Id" headers to the "Composite Key" box, with the word "unique" written next to each arrow.
- The values "121" and "333" in the "Customer Id" column are underlined.
- The values "9446" and "9446" in the "Product Id" column are circled.
- The value "9034" in the "Product Id" column is crossed out with a large 'X' and the word "unique" is written next to it.
- The value "223" in the "Customer Id" column is crossed out with a large 'X'.
- The value "9023" in the "Product Id" column is crossed out with a large 'X'.



Topic : Alternate key

Alternative key is the key in the database table which is not selected as a primary key. So it's an additional key similar to the primary key with which we can access data in the table

Characteristics of Alternative Key

- **Uniqueness**: An alternate key must be unique, which means that no two elements can have the same values. For an alternate key this is important because it ensures that each record of the table can be uniquely identified by the Alternately key Which makes the alternate key ideal as a candidate key and the primary key.
- **Not-Null constraint**: Because an alternate key has the potential to be a primary key. ~~So alternate key cannot~~ have null values. This means that each record in the table must have a value for the alternate key columns. This is because null records cannot be identified and may cause problems with the Integrity of the database.
- **Candidate key**: Candidate key means that the alternate key is a possible candidate for becoming our primary key. If the chosen primary key has some problems and or does not fulfill requirements. An alternate key has all the properties that are necessary to serve as the primary key. However, due to design constraints Or decisions, it serves as an alternative.



Topic : Alternate key

- Student_ID: A unique identifier for each student.
- Email: The student's email address.
- Phone_Number: The student's phone number.

Student_ID	Email	Phone_Number
1	Geeks@gmail.com	88982192
2	Geek@geek.com	37117239
3	Randompeople@gmail.com	89132198

In this table, both Email and Phone_Number could serve as candidate keys because they are unique to each student. If Student_ID is chosen as the Primary Key, then Email and Phone_Number become Alternate Keys. They can be used to uniquely identify a student if the Student_ID is not known.



Topic : Relational Model

In Database Management Systems (DBMS), the relational model organizes data into tables, also known as relations, where each row represents an entity and each column represents an attribute of that entity

Relational Model concept

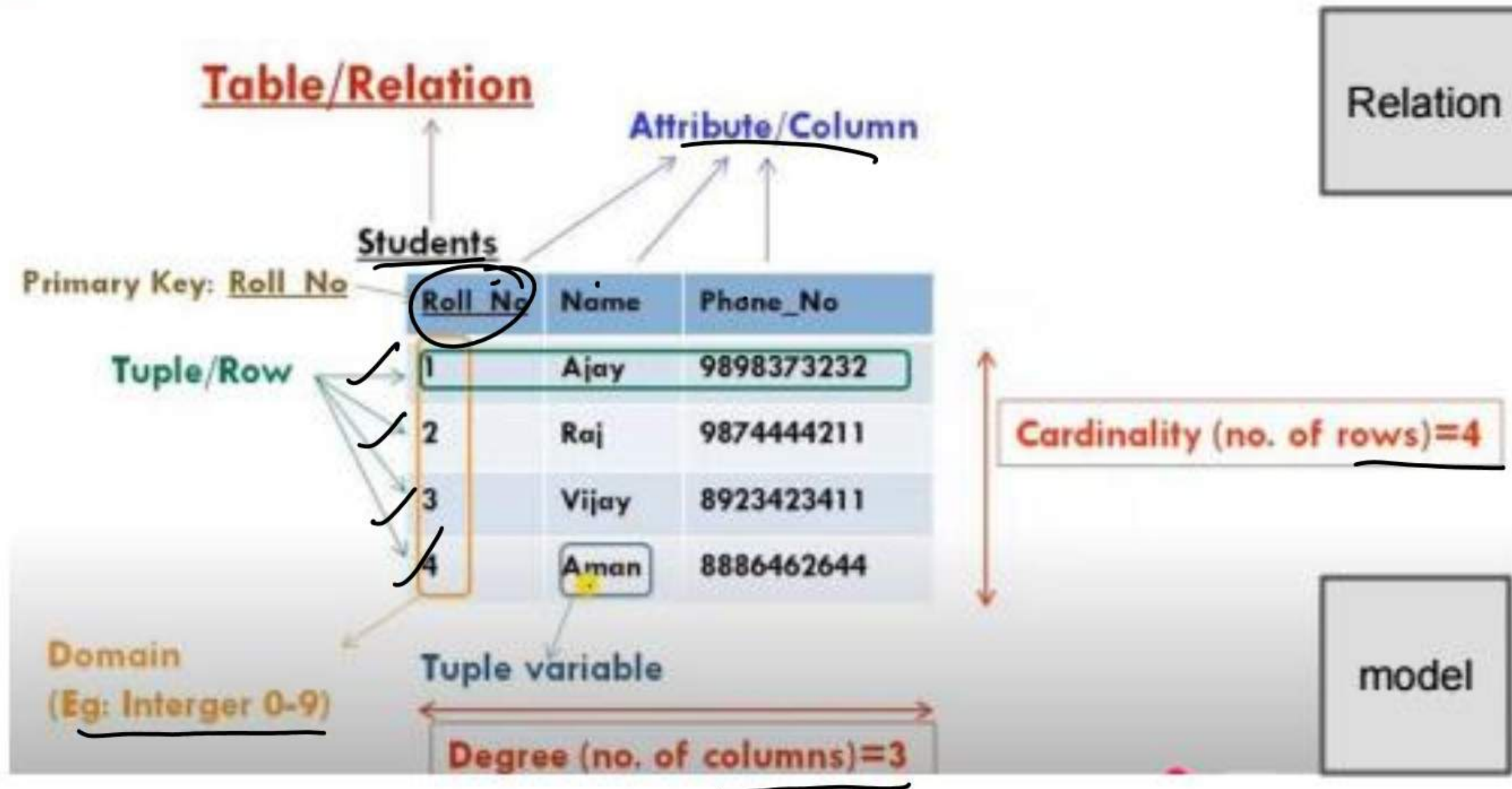
- Relation:** A relation is a table with columns and rows.
- Attribute:** An attribute is a named column of a relation.
- Domain:** A domain is the set of allowable values for one or more attributes. ✓
- Tuple:** A tuple is a row of a relation.
- Relation Schema** ^(table): A relation schema represents the name of the relation with its attributes.
- Relation instance** (State): Relation instance is a finite set of tuples. Relation instances never have duplicate tuples.

Roll No
Number
of
*

Roll_No	Name	Phone no
1	Ajay	9898373232
2	Raj	9874444211
3	Vijay	8923423411
4	Aman	8886462644



Topic : Relational Model



Degree:- The Number of Attributes(column) in relation.
Cardinality:- The number of Tuples (rows) in a relation.



Topic : MCQ

1. A view of database that appears to an application program is known as
✓ (A) Schema _(B) Subschema _(C) Virtual table _(D) Index Table
2. What is rows of a relation known as?
(A) Degree _(B) Entity _(C) Tuple _(D) None
3. Which is the lowest level of abstraction that describes how the data are actually stored?
(A) Physical _(B) Abstract _(C) View _(D) User
4. Which of the following allows to uniquely identify a tuple?
✓ (A) SCHEMA _(B) Attribute _(C) SUPER KEY _(D) Domain
5. Which of the following is known as minimal super key?
(A) Primary key _(B) Foreign Key _(C) Candidate Key _(D) Alternate key
6. How many levels are there in architecture of database?
(A) 1 _(B) 3 _(C) 2 _(D) 5



Topic : MCQ

7. Primary key can be?

- (A) NULL (B) ~~NOT NULL~~ (C) BOTH A and B (D) OPTION (D) Depend situation

8. Select the correct advantages of view.

- (A) Data Security (B) Hiding for Complex Queries (C) none of these
(D) ~~All of these~~

9. Select the correct foreign key constraint?

- (A) ~~Referential Integrity~~ (b) Entity Integrity (c) Domain Integrity
(D) None of these

10. Select the correct properties of entities?

- (A) Table (B) Groups (C) ~~Attributes~~ (D) None of these



Topic : MCQ

10. Select the definition of the correct key which is used to represent relation between two tables?

- (a) Candidate key (b) Foreign key (c) Primary key (d) Super key

11. Total view of a database is known as?

- (a) Physical view (b) Internal View (c) Conceptual View (d) None of these

12. Which of the following is NOT a characteristic of a DBMS?

- (a) Data redundancy (b) Data integrity (c) Data security
(d) Data independence

13. The process of organizing data into tables and establishing relationships between them is known as:

- (A) Data mining (B) Data modelling (C) Data warehousing

(D) Database design



Topic : MCQ

14. Which of the following is NOT a type of integrity constraint in a relational database?

- (a) Referential integrity (B) Semantic integrity (C) Structural integrity
(D) Domain integrity

15. An attribute that uniquely identifies each entity in an entity set is called a:

- (A) Composite attribute (B) Derived attribute (C) Multi-valued attribute
(D) Key attribute

16. For each attribute of relation, there is a set of permitted values called the _____ of that attribute.

- (A) Domain (B) Relation (C) Set (D) Schema

17. Consider A Relationship Schema $R(A, B, C)$ Where A is Candidate Key. Then Total number of Super Key Possible ?

- (A) 2 (B) 4 (C) 6 (D) 8

No. Super Key = 2^{n-1} , n = total no of Attribute
 $= 2^{3-1} \Rightarrow 2^2 \Rightarrow 4$
 $1 \rightarrow A$
 $2 \rightarrow AB$
 $3 \rightarrow AC$
 $4 \rightarrow ABC$



Topic : MCQ

18.

Consider the following statements S1 and S2 about the relational data model:

S1: A relation scheme can have at most one foreign key.

X

S2: A foreign key in a relation scheme R cannot be used to refer to tuples of R.

Which one of the following choices is correct?

1. S1 is true and S2 is false.

2. Both S1 and S2 are true.

3. ☒ Both S1 and S2 are false.

4. S1 is false and S2 is true.



Topic : Discussion Some Crucial type MCQ

MCQ DISCUSSION

1. MCQ1:- A view of database that appears to an application program is known as

(A) Schema (B) Subschema (C) Virtual table (D) Index Table

DISCUSSION:- A **subschema** is a subset of the schema and **inherits the same property** that a **schema has**. The **plan for a view** is often called **subschema**. Subschema refers to an application programmer's view of the data item types and record types, which he or she uses. Option (B) is correct.

2. MCQ 12:- Data redundancy is not a characteristic of a DBMS; rather, DBMS aims to minimize data redundancy through normalization and other techniques to ensure data is stored efficiently.



Topic : Discussion Some Crucial type MCQ

3. MCQ 14:- Structural integrity is not a type of integrity constraint in relational databases. The primary types include domain integrity, referential integrity (ensuring foreign keys correctly refer to primary keys), and semantic integrity (ensuring data adheres to business rules).
4. MCQ 16:- For Each attribute of a relation, there is a set of permitted values, also known as the domain of that attribute.



Topic : Discussion Some Crucial type MCQ

5. MCQ 17:-

Any superset of a key (primary key or candidate key) is a superkey.

Data:

$n \rightarrow$ total number of attributes = 3

$n(s) \rightarrow$ total number of super keys

Since A is a primary key it must be included in every superkey.

B, C may or may not be included in superkey.

Formula:

Since only one superkey is present in the relation:

$$n(S) = 2^n - 1$$

Calculation:

$$n(S) = 2^{3-1} = 2^2 = 4$$

Superkeys are: A, AB, AC, ABC.



Topic : Discussion Some Crucial type MCQ

6. MCQ 18.

Concept:

Foreign Key : is the set of attributes in a particular relation whose values are belongs to primary key of same relation or other relation.

Explanation:

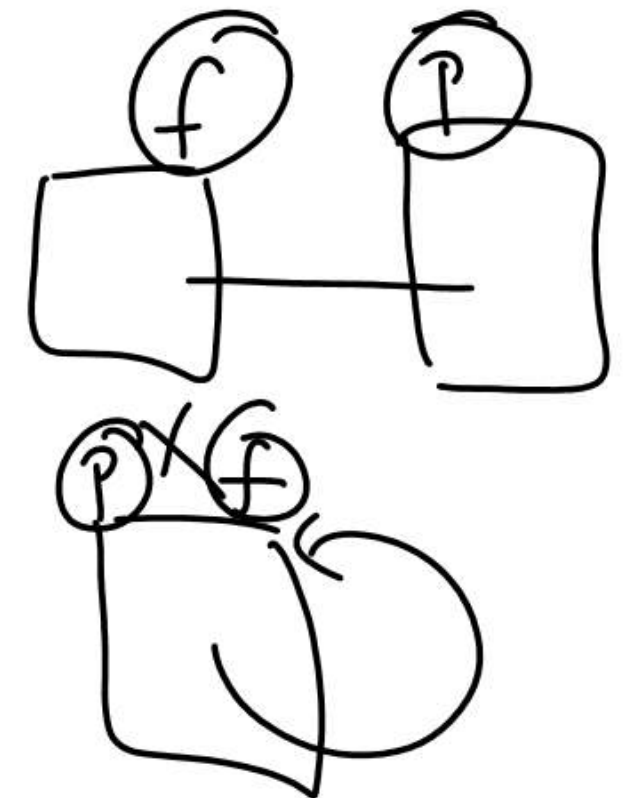
Statement 1: A relation scheme can have at most one foreign key.

There is **no such restriction** on how many number of Foreign keys a particular relation can have. A relation can have **as many number of Foreign keys as Required**.

So this statement is **false**.

Statement 2: foreign key in a relation scheme R cannot be used to refer to tuples of R.

There is no such constraint. Foreign key can be used to refer to primary key of the same relation. Self-referencing relations are examples of such foreign key. So this statement is also **false**.





Topic : Recap

- 1:- Super Key ✓
- 2:- Candidate Key ✓
- 3:- Primary key ✓
- 4:- Foreign Key ✓
- 5:- Composite Key ✓
- 6:- Alternate Key ✓
- 7:- Relational Model ✓
- 8:- MCQ Problem solving ✓
- 9:- Discussion on MCQ ✓

THANK YOU